

HANDS - ON RAG BOT & AGENTIC AI



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RAG DEMO

- <https://huggingface.co/spaces/rajeshboppana/EcoTourismBot>
- what is the top season which has more visitor

WHAT WE ARE GOING TO DO TODAY?

Steps:

Data Preparation: Load and process the dataset using pandas

Vector Store Setup: Convert product descriptions into embeddings.

Store embeddings in a vector database:

Building the Chatbot:

Use LangChain to create an LLM pipeline.

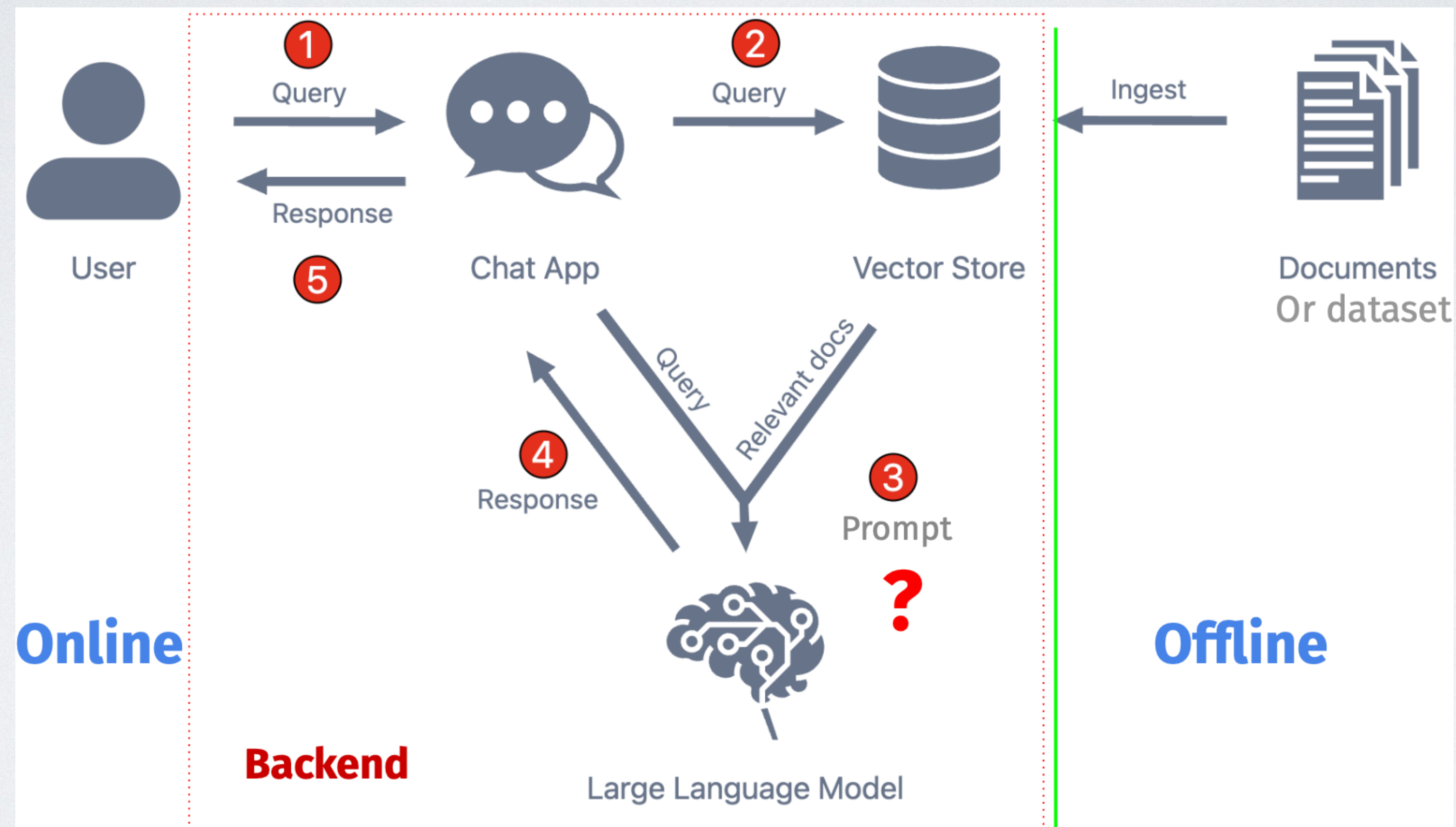
Develop a simple chatbot to answer EcoTourism related queries.

Creating a UI: Implement a Gradio-based UI for user interaction.

WHY RAG IS IMPORTANT

RAG is crucial because LLM models, like ChatGPT, can sometimes "hallucinate" or provide outdated or incorrect information. By retrieving facts from trusted sources before generating responses, RAG ensures the answers are more reliable, up-to-date, and contextually relevant.

RAG ARCHITECTURE



RAG Evaluation

Metrics:

- a. Helpfulness
- b. Completeness
- c. Correctness

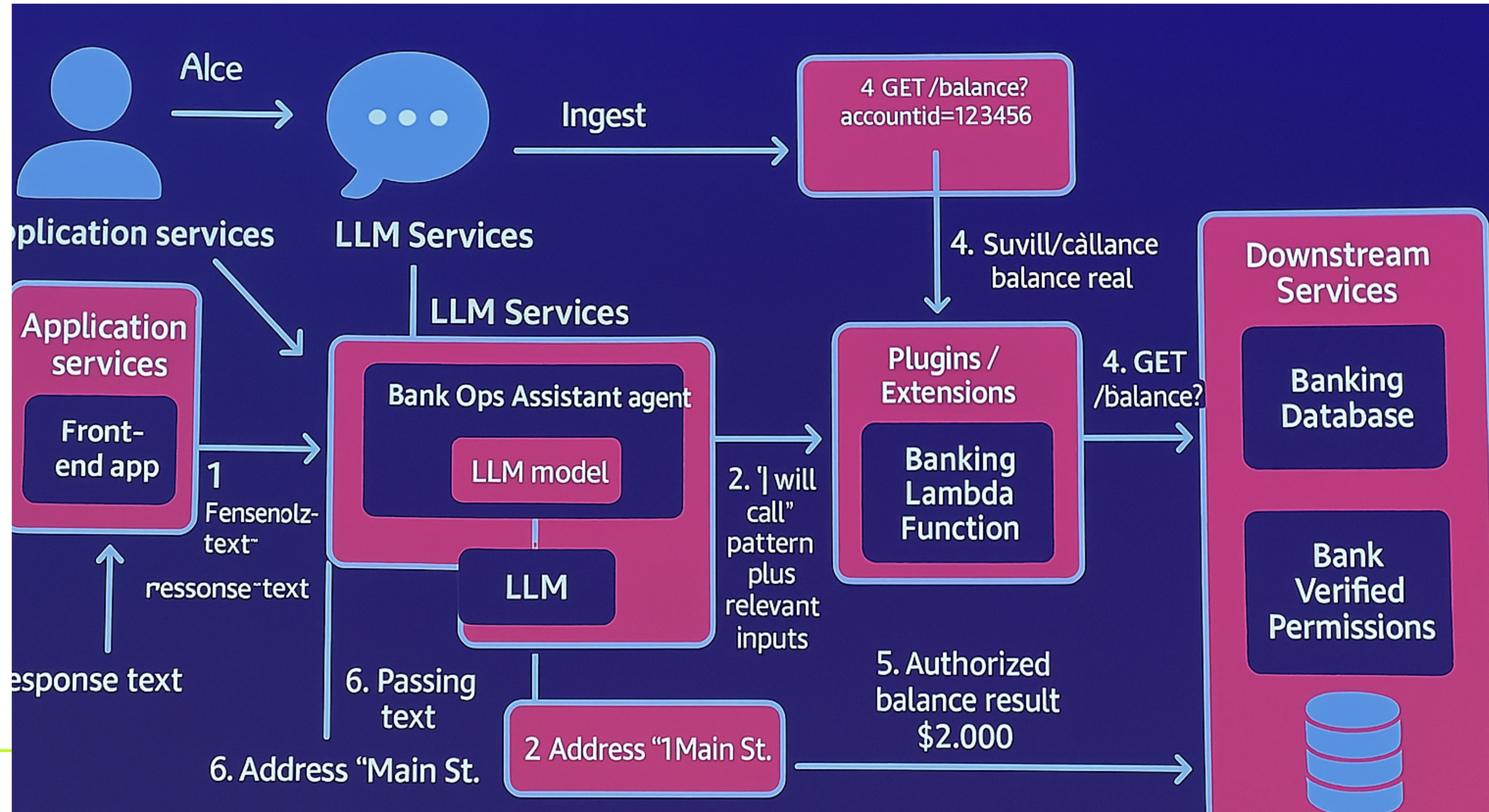
Responsible AI

- a. Harmfulness

Steps:

1. Setting up the Knowledge Base
2. Uploading & Configuring data
3. Creating Golden Data Set
4. Running evaluations & Metrics
5. Reviewing evaluation Results

How to Secure RAG/ OWASP Principles



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Questions ??

AGENTIC AI DEMO

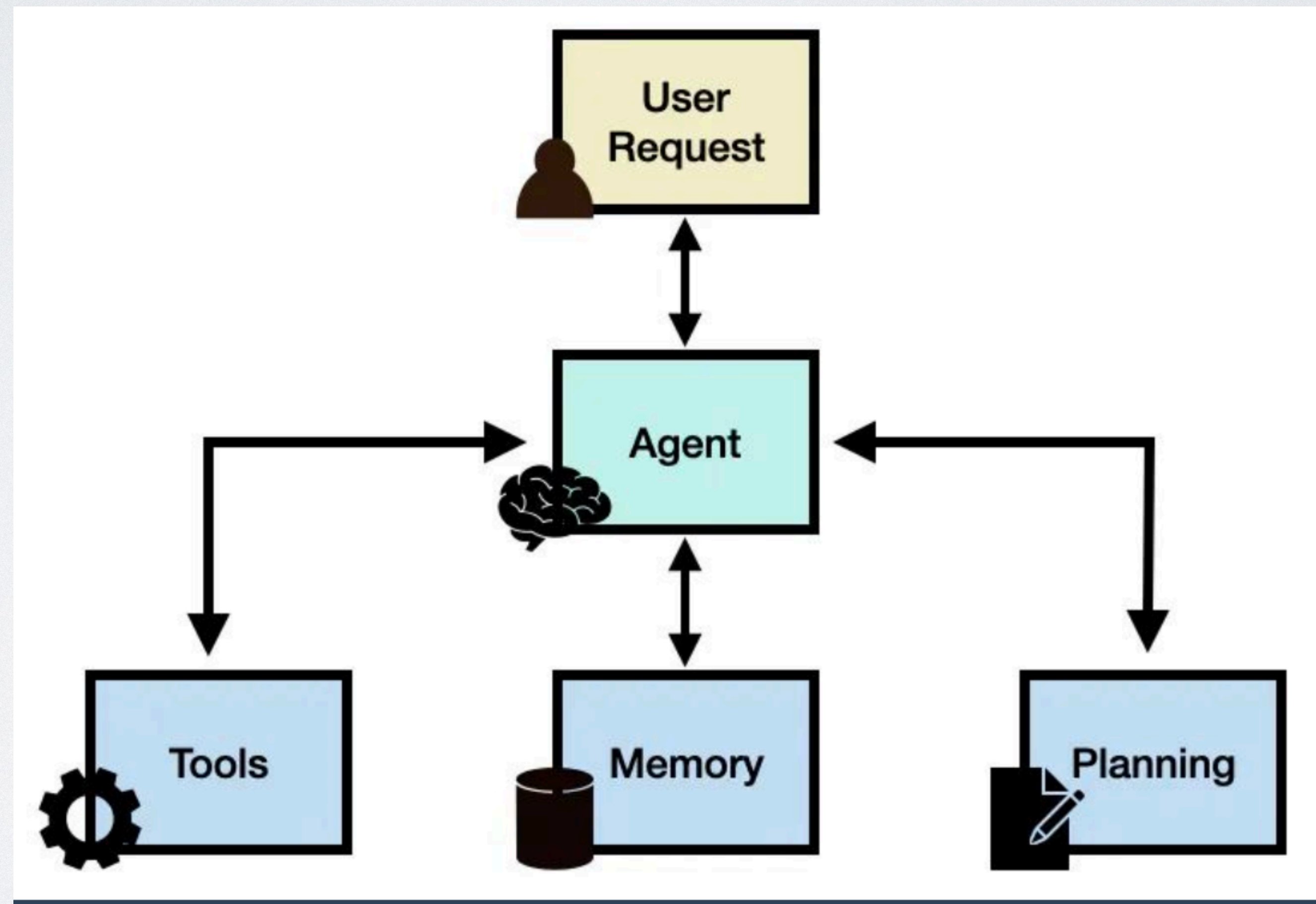
- Book a Doctor appointment with Dr. lee at 3pm?

WHY WE NEED AN AGENT

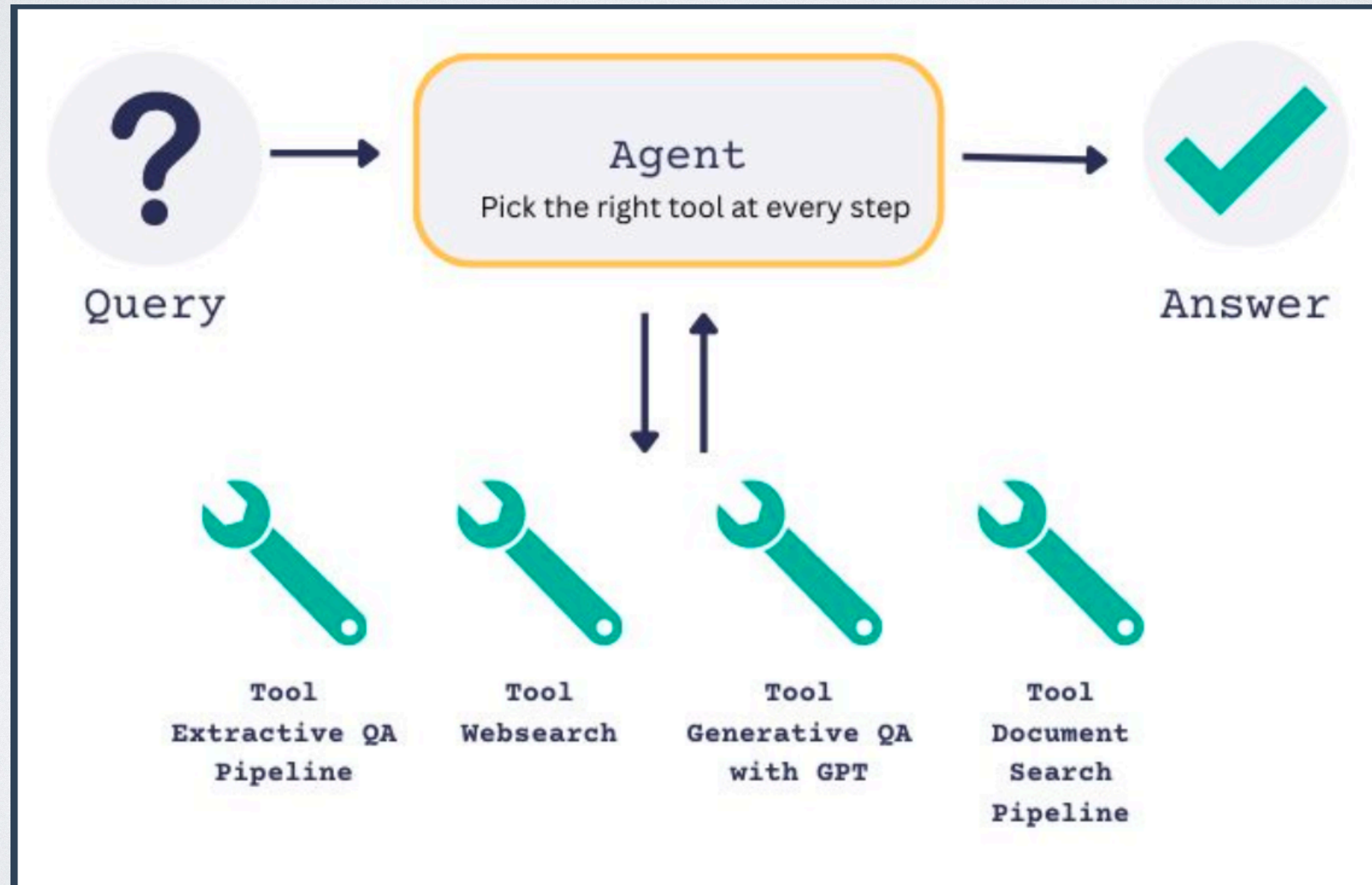
Enabling complex task execution, going beyond simple text generation by allowing AI systems to not only understand language but also:

- **Take Actions**
- **Make Decisions**
- **Interact with the real world.**

AGENT ARCHITECTURE



VARIOUS TYPES OF TOOLS THAT AGENT CAN CONNECT



Questions ??
